

## Module 6.5

# Cybersecurity and Drone Security



### Executive Abstract

The contemporary landscape of Unmanned Aerial Vehicles (UAVs) has transitioned from rudimentary remote-controlled platforms to sophisticated, autonomous nodes within the Internet of Unmanned Aerial Vehicles (IoUAV). This transition dictates a commensurate shift from perimeter defense to Zero Trust Architecture.

### System Telemetry

**Protocol:**

MAVLink (Unencrypted Default)

**Positioning:**

GNSS (Susceptible at -128 dBm)

**Hardware Trust:**

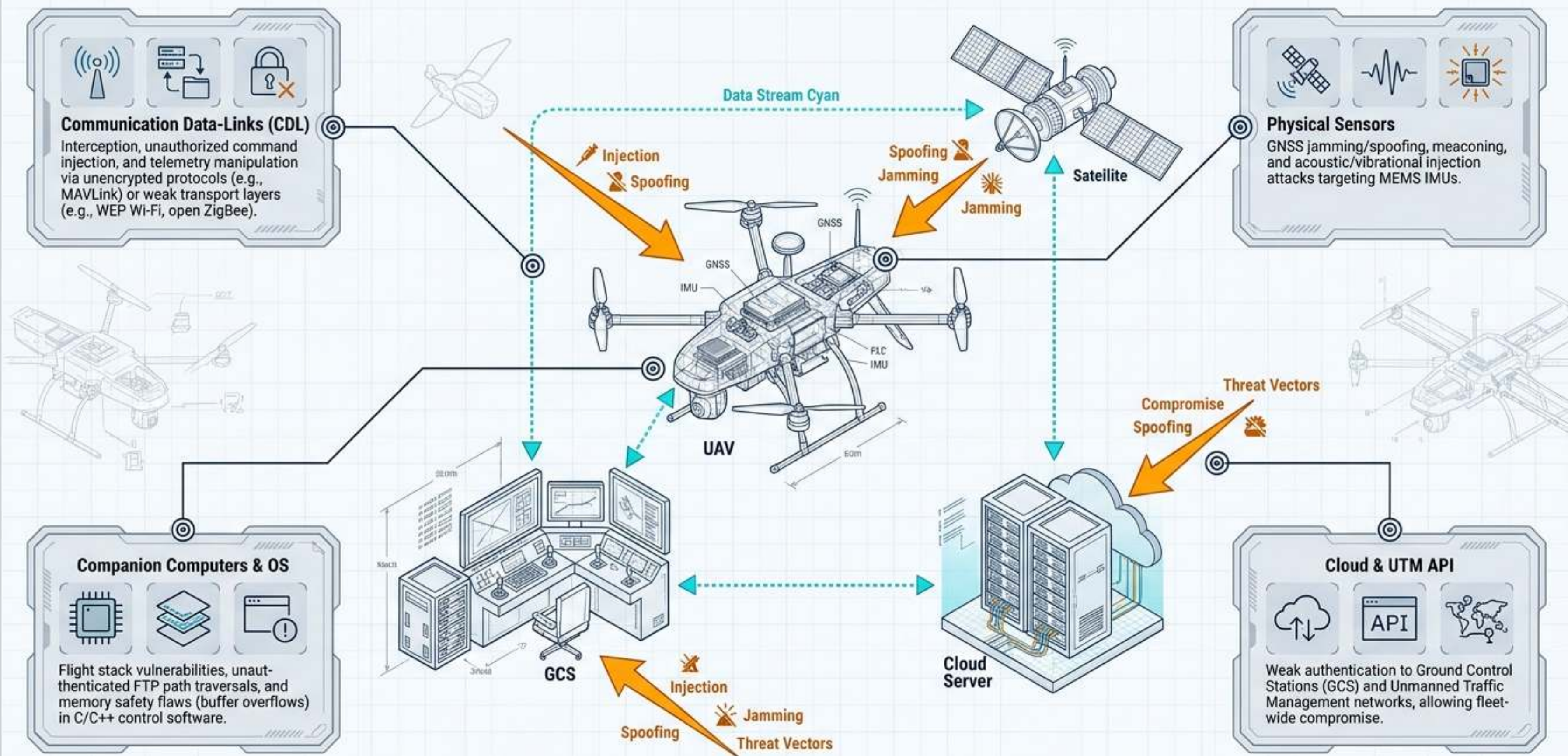
TEE / HSM / TPM / SE required.

**Threat Surface:**

Physical, Network, Transport, and Application Layers.



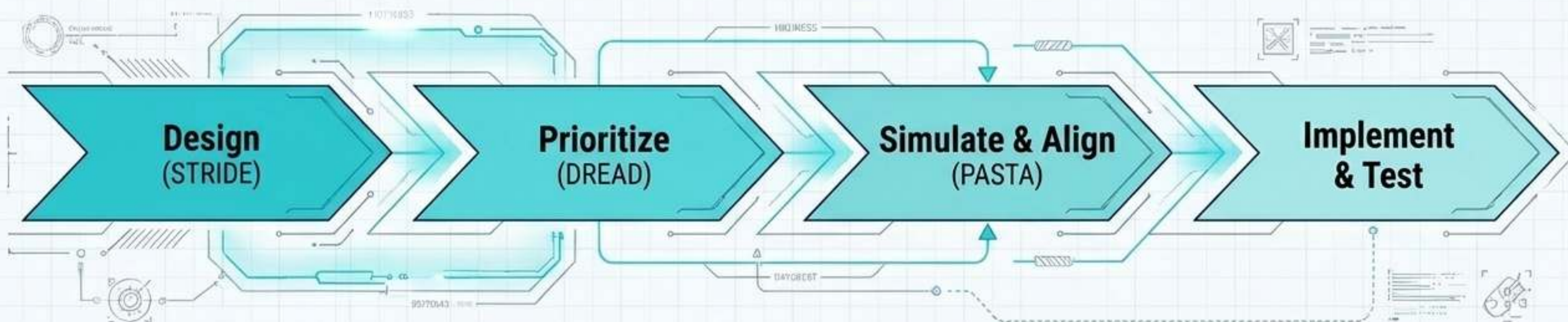
## The Systemic Threat Map of the IoUAV Ecosystem





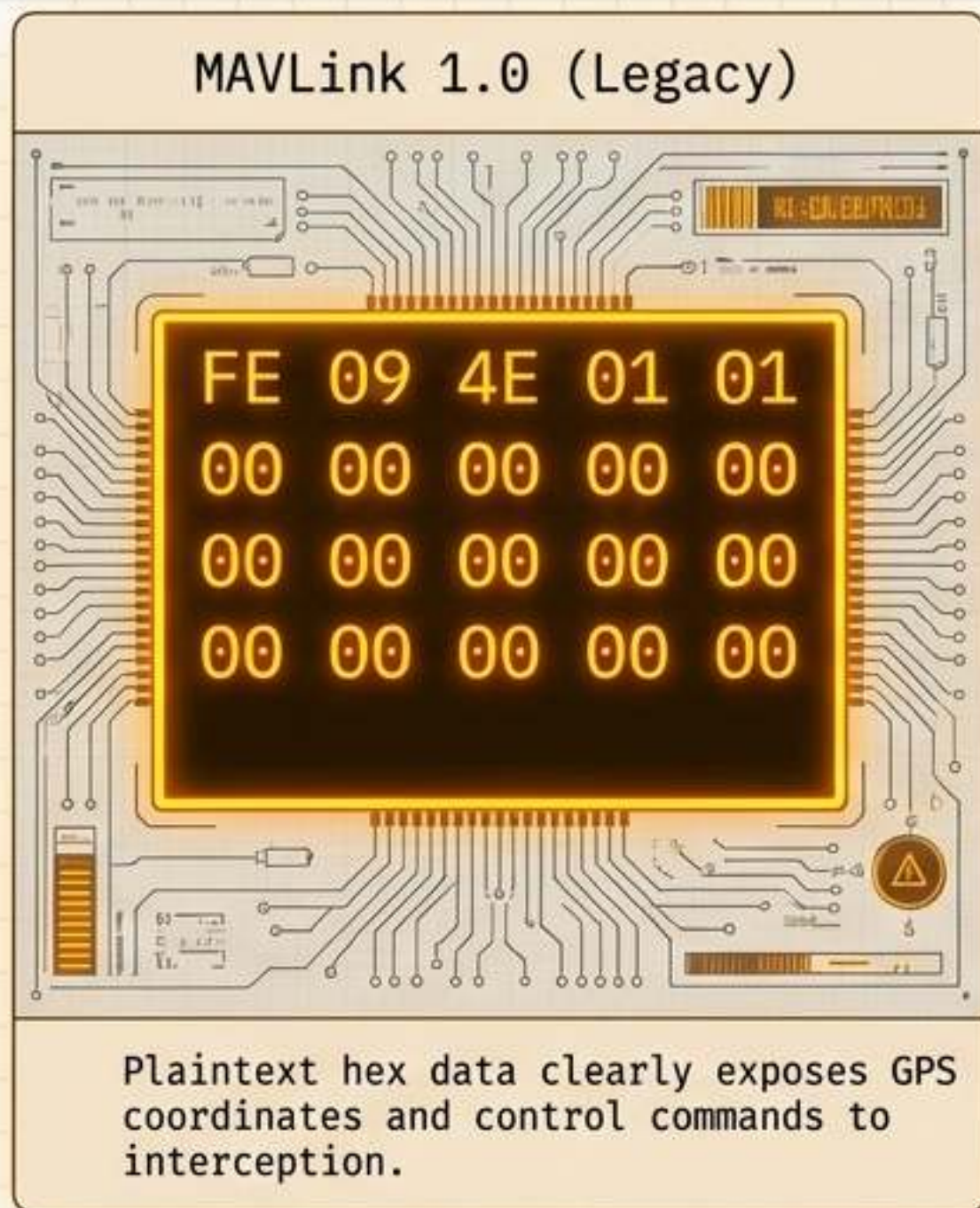
## Integrating Threat Modeling Frameworks into UAV Design

| STRIDE                                                                                                                                                                                                         | DREAD                                                                                                                                                                                                                                         | PASTA                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identification                                                                                                                                                                                                 | Quantification                                                                                                                                                                                                                                | Business Alignment                                                                                                                                                                                                                             |
| <p>Applied during the design phase to identify threats tied to system properties.</p> <p>Analyzes Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.</p> | <p>Used alongside STRIDE to score and rank threats.</p> <p>Evaluates Damage potential, Reproducibility, Exploitability, Affected Users, and Discoverability (e.g., scoring an unauthenticated RCE higher than a physical-access exploit).</p> | <p>A 7-step Process for Attack Simulation and Threat Analysis.</p> <p>Used for enterprise-level fleet operations to map technical vulnerabilities (e.g., loss of C2 link) to operational impact (e.g., payload theft, airspace violation).</p> |

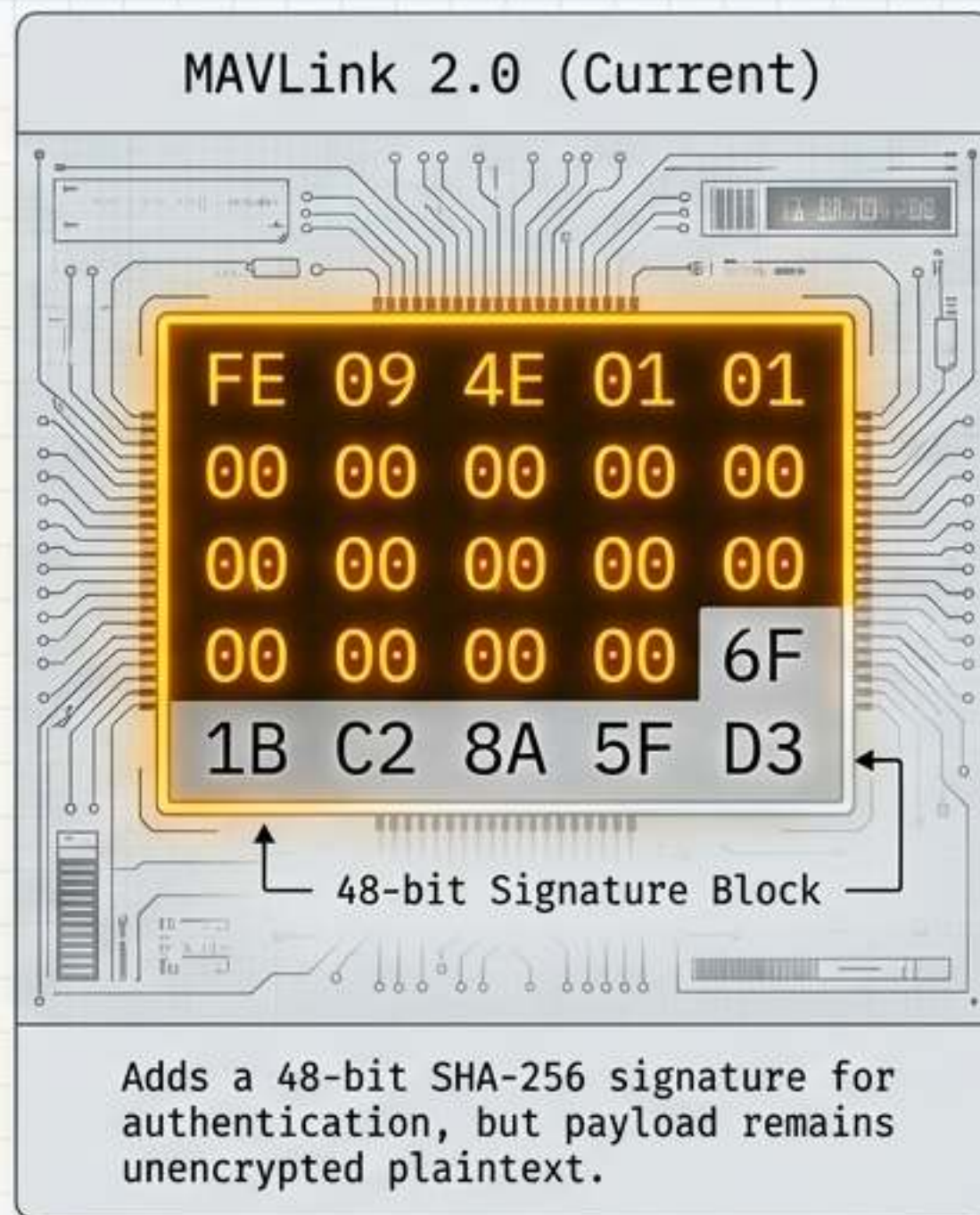




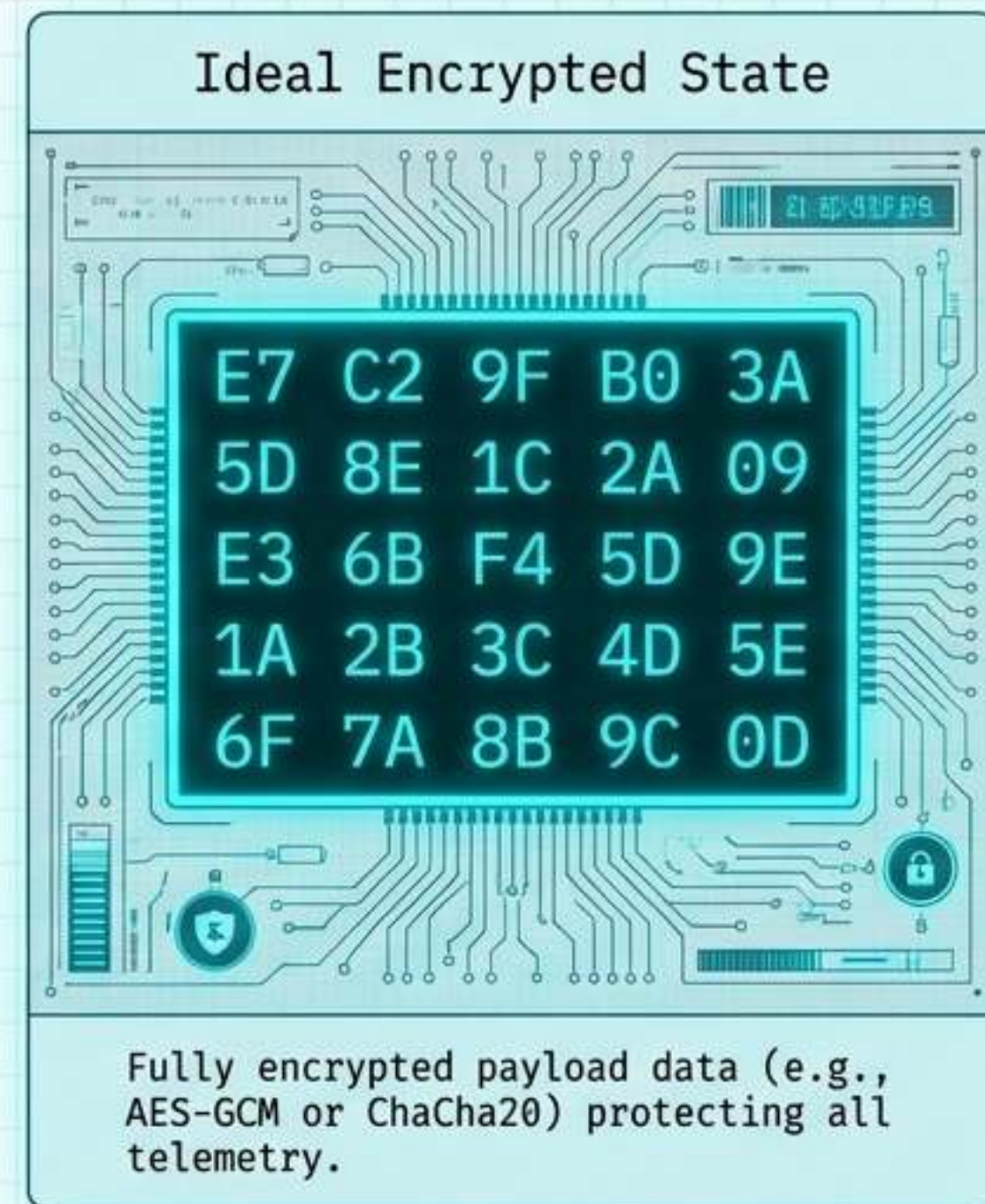
# Protocol Vulnerabilities and the MAVLink Plaintext Problem



**⚠️ Eavesdropping:** Telemetry exposes mission parameters and edge AI models to SDR interception.



**⚠️ Command Injection:** Without mandated message signing, attackers forge packets to hijack flight paths.



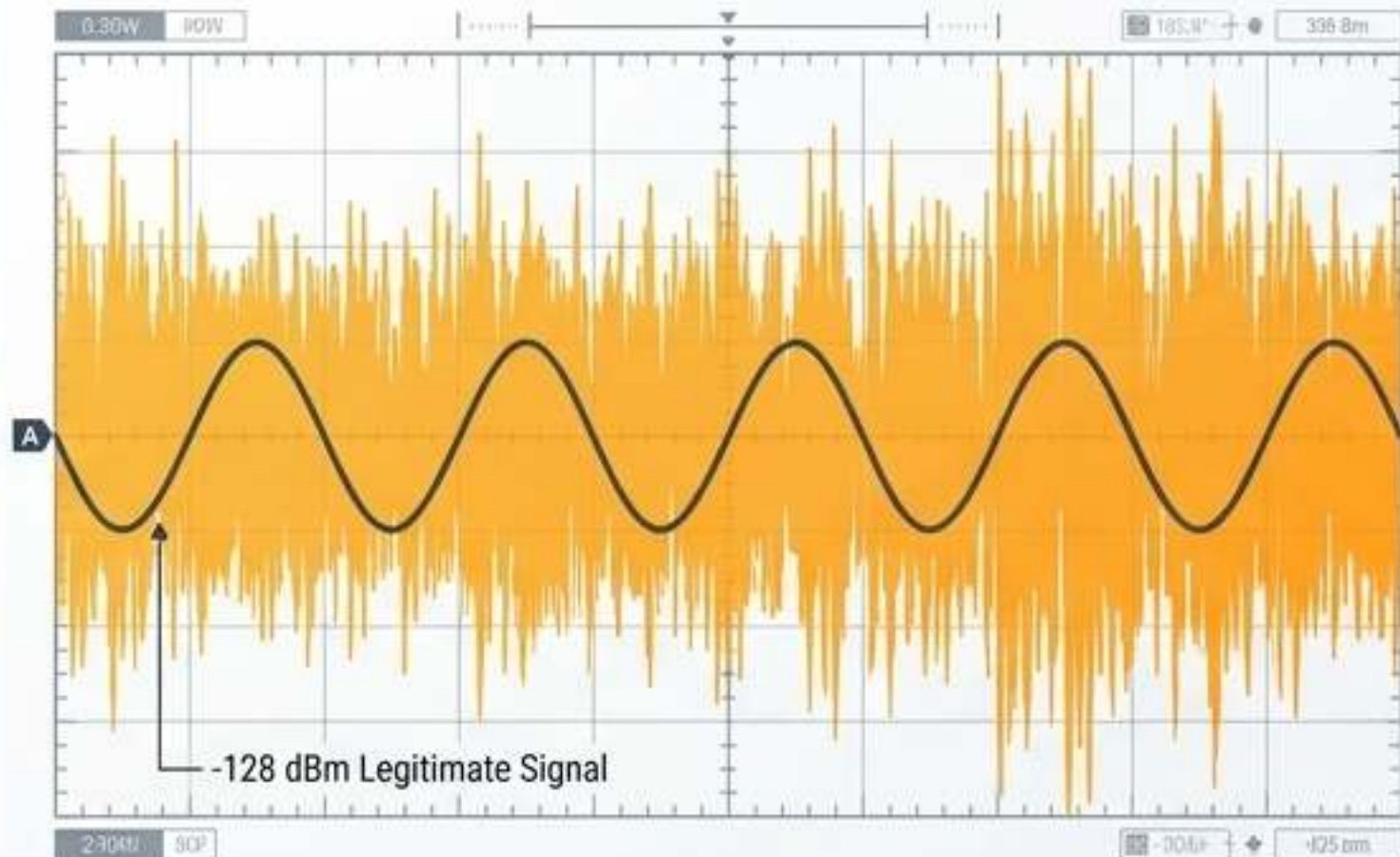
**🛡️ MitM & Heartbeat Exploits:** Attackers intercept unencrypted channels to manipulate critical 1-byte heartbeat messages.



# Electronic Warfare: The Physics of GNSS Interference

**A**

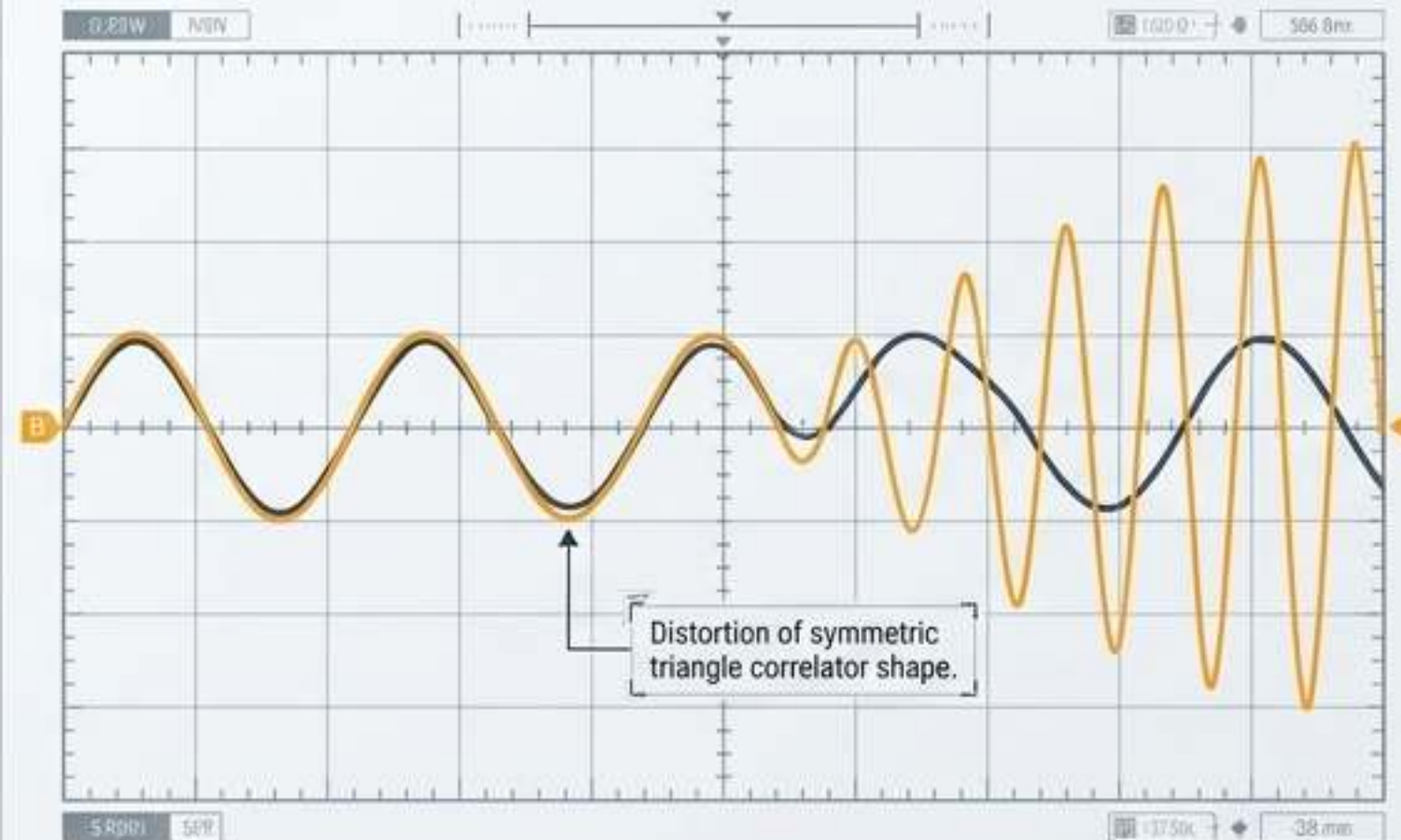
## Brute-Force Jamming



Overpowers civilian bands (L1/L2/L5), forcing emergency fail-safes (hover/RTH). Effectiveness measured by the J/S ratio.

**B**

## Generative Spoofing



Transmitting synchronized, counterfeit signals to secretly hijack navigation. Detectable via Doppler Shifts and Correlator Shape distortion (code delay within  $\pm 1$  chip).



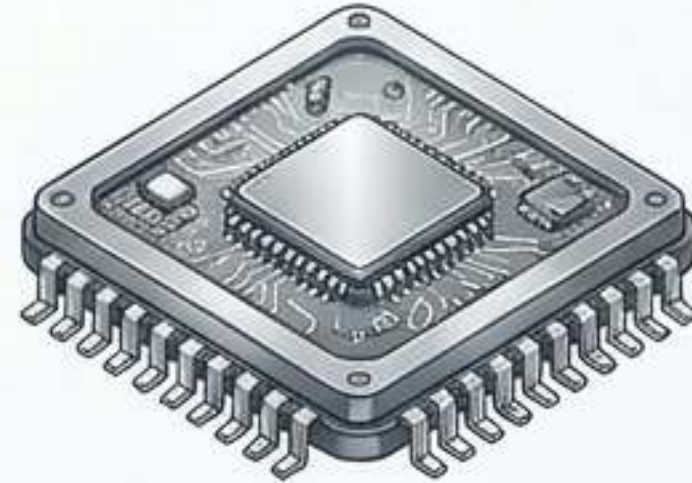
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## Forwarding (Meaconing)

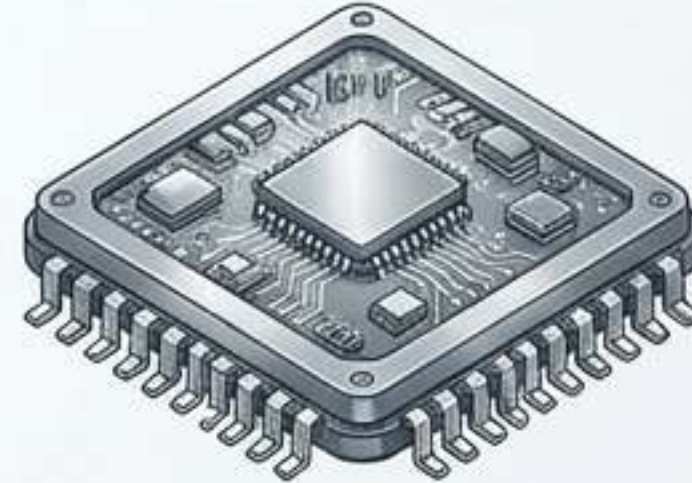
Capturing and retransmitting genuine satellite signals with an amplified time delay. Detectable via Time Stamp Anomalies.







## Secure Element (SE)

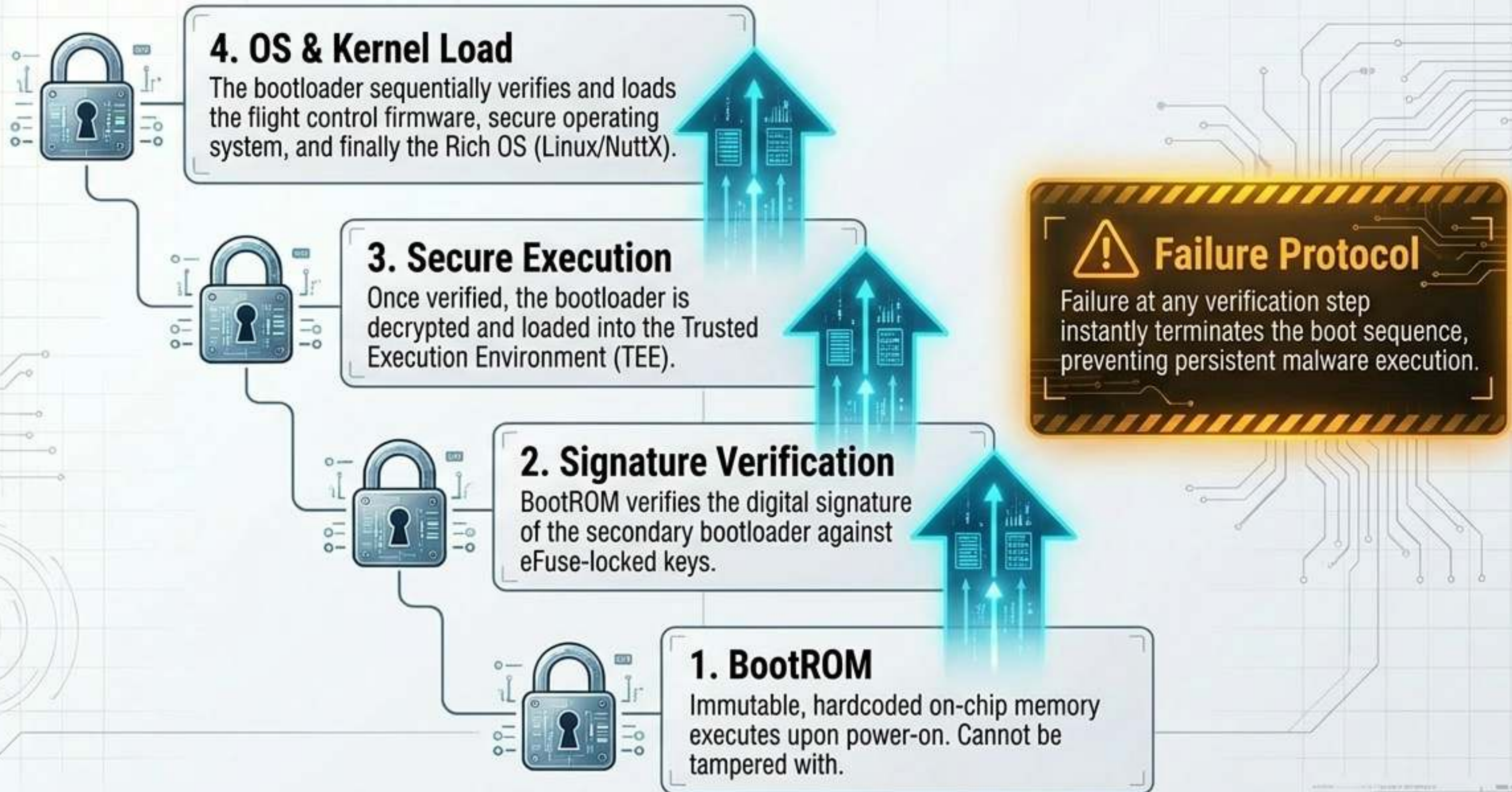


A detailed illustration of a square integrated circuit (chip) with a central square die, surrounded by a grid of pins and various surface components.

**Use Case:**  
Flight controller  
isolation.



# Cryptographic Integrity and The Secure Boot Chain





# Enterprise Application Hardening and Data Privacy



## OS-Level Defense (SELinux)

Enforces mandatory access control policies for all processes, operations, and files, stored in the root filesystem and protected by DM-Verity.

## Runtime Protection

Employs dynamic code obfuscation, adding redundant logic to thwart reverse-engineering. White-box cryptography protects keys.

## Data Isolation (Local Data Mode)

An essential privacy control that entirely disconnects the flight application from the internet, preventing sensitive telemetry transmission.

## At-Rest & In-Transit Security

User data is secured via AES-256 encryption in storage, while TLS secures all cloud transmissions.



# Memory Safety Exploits within the Flight Stack

## CVE-2026-32706 (crsf\_rc parser)



**CVSS 8.1:** Parser accepts an oversized, variable-length packet without bounds checking. An adjacent attacker can crash the autopilot causing total loss of flight control.

## CVE-2026-32743 (MavlinkLogHandler)



**CVSS 6.5:** Stack-based overflow triggered via MAVLink FTP log requests. The sscanf function parses paths without a width specifier, crashing the MAVLink task and causing DoS.

### Failure Protocol

**CRITICAL FAILURE:** Instantaneous termination of compromised process, preventing escalating impact.



# Applied Cryptography: Overcoming MAVLink Overhead



## The Encryption Dilemma

Standard MAVLink eschews encryption to preserve low latency. AES-CTR, ChaCha20, Speck-CTR, and Rabbit ciphers were evaluated on live PX4/ArduPilot testbeds.

## The MAVShield Cipher Architecture



Proven robust against differential cryptanalysis, passing 24 distinct NIST/Diehard attacks while providing comprehensive confidentiality with negligible overhead.



# Cyber-Physical Exploits: Acoustic and Inertial Attacks

## Resonant Frequency Injection

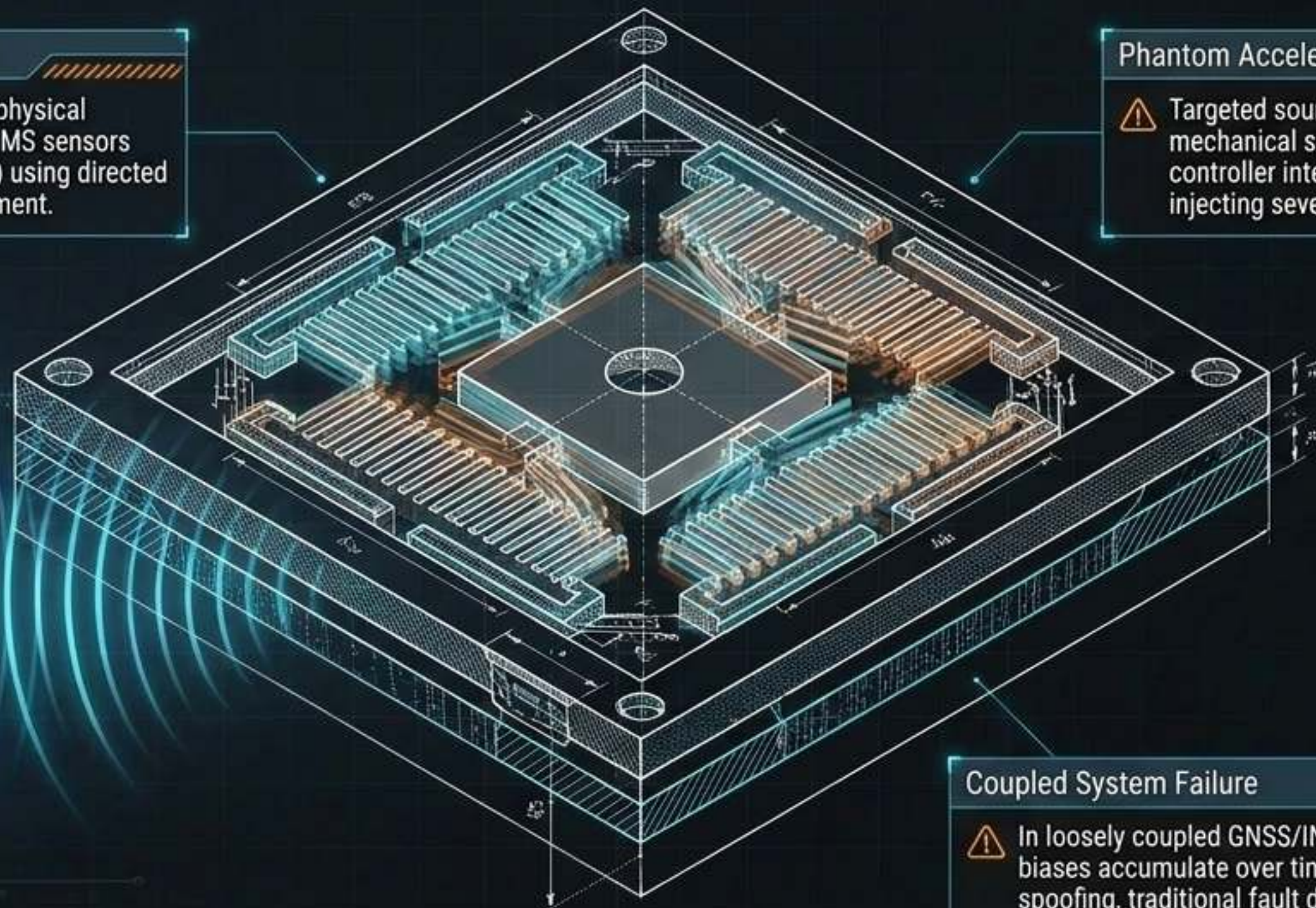
⚠️ Attackers target the specific physical resonance frequencies of MEMS sensors (typically between 15-25 kHz) using directed acoustic or vibrational equipment.

## Phantom Acceleration

⚠️ Targeted soundwaves induce false mechanical signals that the drone's flight controller interprets as actual movement, injecting severe angular-rate biases.

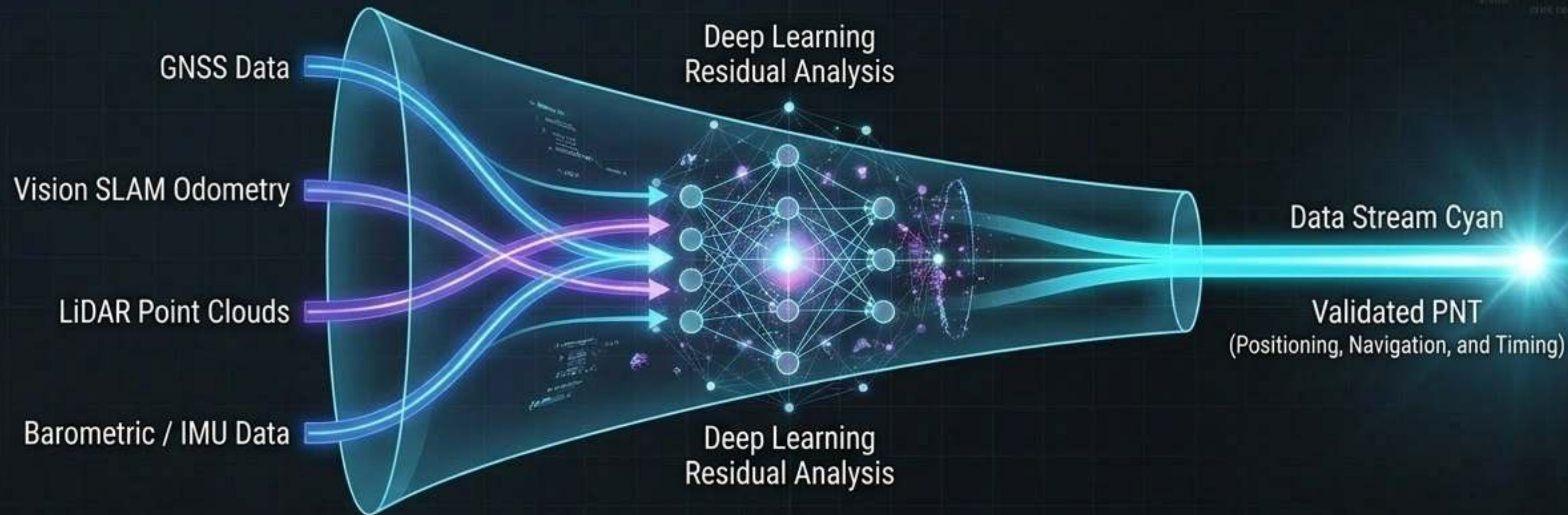
## Coupled System Failure

⚠️ In loosely coupled GNSS/INS architectures, these injected biases accumulate over time. If synchronized with GNSS spoofing, traditional fault detection mechanisms collapse, allowing seamless trajectory hijacking.





# Defeating Spoofing via AI-Assisted Sensor Fusion



## Cross-Verification

Relieving exclusive reliance on GNSS by cross-referencing inertial data with visual odometry and 5G positioning.

## Innovation Sequence

Monitoring the difference between predicted and measured values. Statistical hypothesis testing triggers alerts when residuals exceed expected bounds.

## Signal Watermarking

Analyzing the entropy of incoming signals to verify their cryptographic authenticity before integration, defeating sophisticated spoofing.



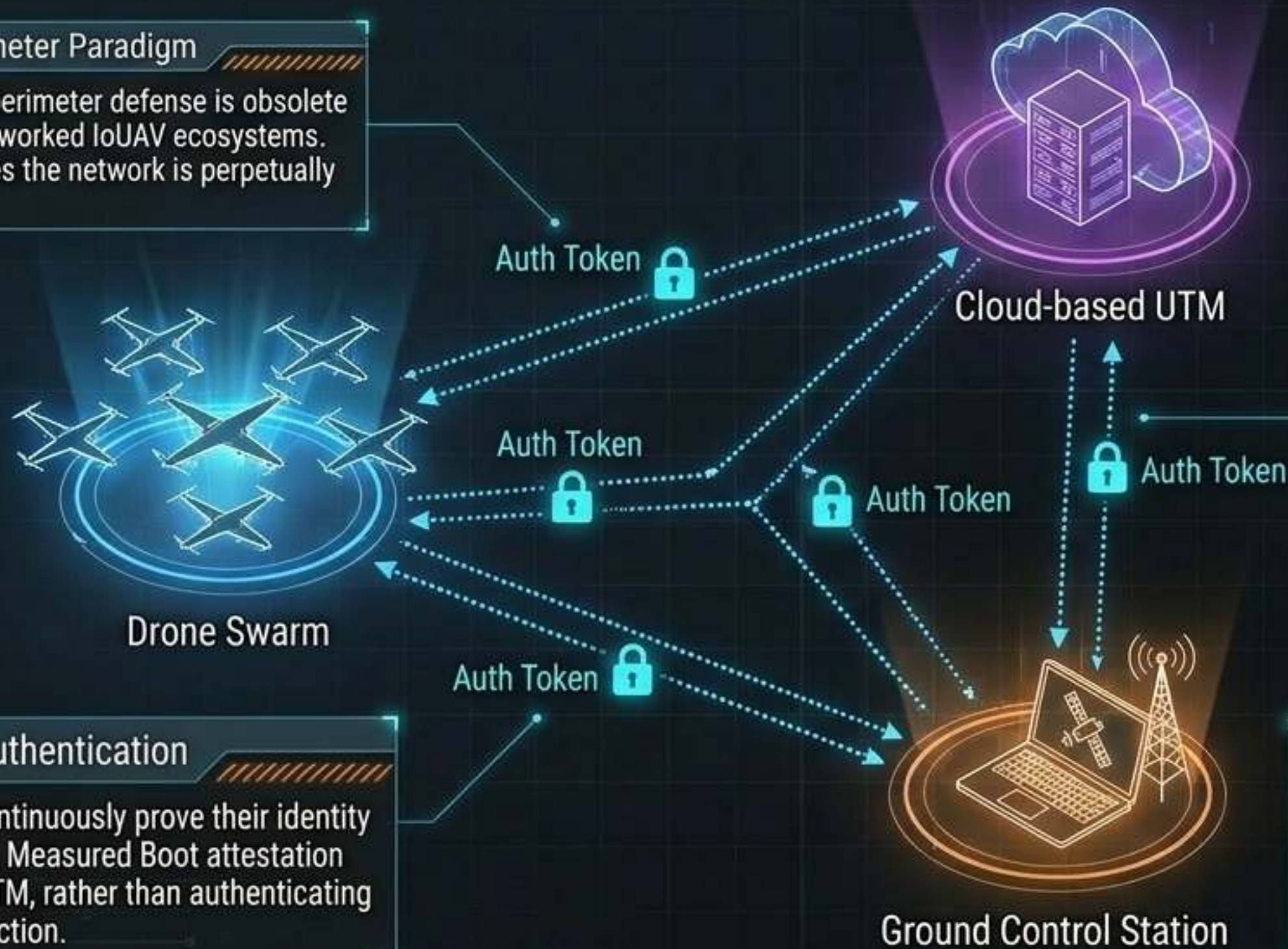
# Zero Trust Architecture in UAS Traffic Management (UTM)

## The Post-Perimeter Paradigm

⚠ Traditional perimeter defense is obsolete in highly networked loUAV ecosystems. ZTA assumes the network is perpetually hostile.

## Micro-segmentation

⚠ Restricting lateral movement. A compromised companion computer cannot access the flight controller's PID loop without re-authentication.



## Continuous Authentication

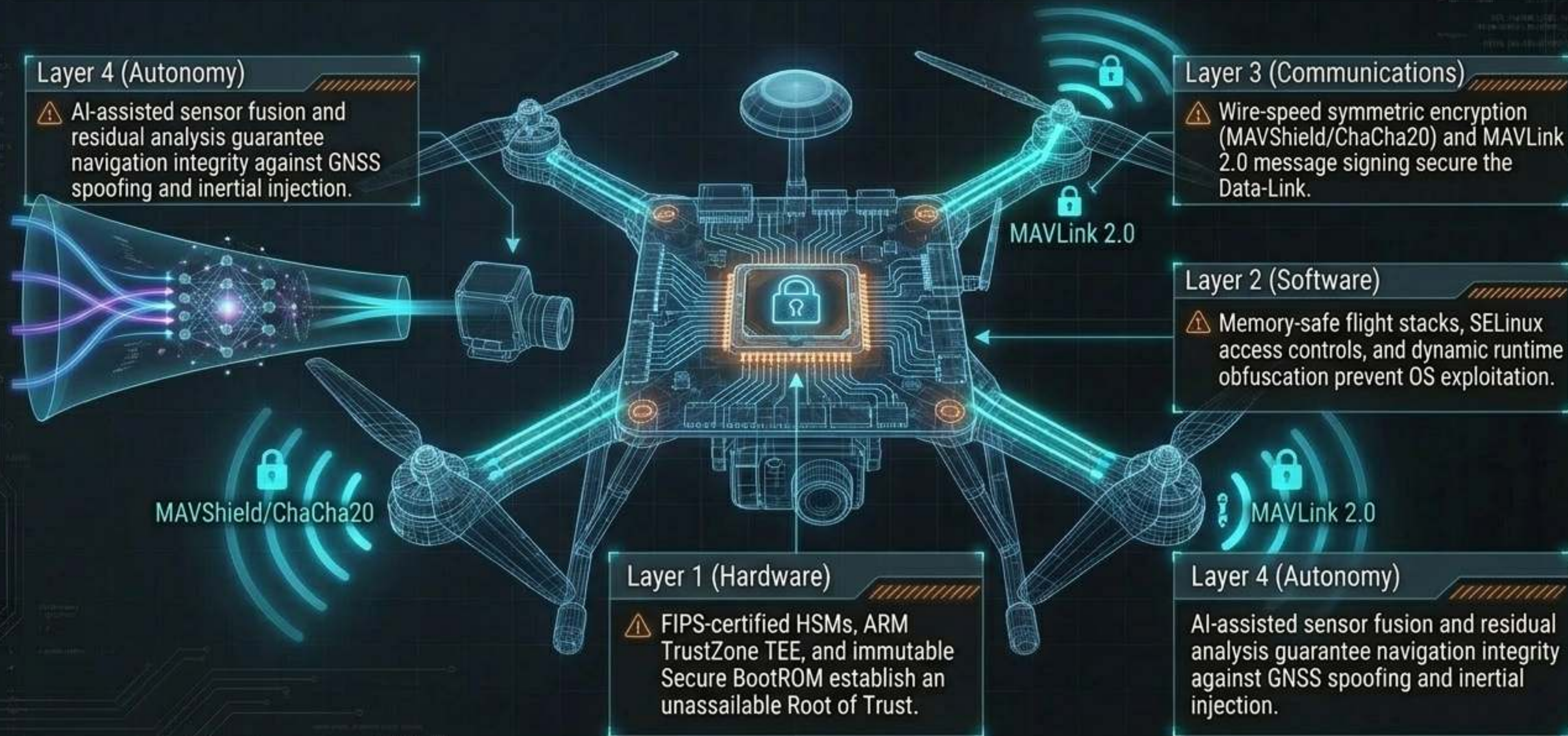
Devices must continuously prove their identity and integrity (via Measured Boot attestation tokens) to the UTM, rather than authenticating only upon connection.

## Dynamic Trust Monitoring

⚠ Ground stations evaluate drone behavior (trajectory, energy use) in real-time to calculate dynamic trust scores, instantly isolating anomalous nodes.



# Synthesis: The Complete Defense-in-Depth Paradigm





# The Future of IoUAV Engineering



**As Unmanned Aerial Systems evolve into critical national infrastructure, cybersecurity can no longer be an overlay—it must be a foundational engineering prerequisite.**

## The Mandate

Future deployment requires mandatory hardware acceleration for cryptographic workloads, standardized post-quantum resilience, and physical tamper-proofing across all commercial and defense platforms.

## Reference

Data and benchmarks derived from recent architectural analyses of PX4 Autopilot CVEs, MAVSec implementations, and DJI Enterprise White Paper specifications.